9. If $3\tan\theta = 4$, then $\sqrt{\dfrac{1-\sin\theta}{1+\sin\theta}}$ is equal to

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) None of these

10. If $\tan A = 1$ and $\tan B = \sqrt{3}$, then $\cos A \cdot \cos B - \sin A \cdot \sin B$ is equal to

(a) $\frac{1+\sqrt{3}}{2\sqrt{2}}$ (b) $\frac{1-\sqrt{3}}{2\sqrt{2}}$ (c) $\frac{2\sqrt{2}}{3}$ (d) 1

11. The value of $\operatorname{cosec}^2\theta - 2 + \sin^2\theta$ is always

(a) less than zero (b) non-negative
(c) zero (d) 1

12. What is the value of
$\sin^2 15° + \sin^2 20° + \sin^2 25° + \cdots + \sin^2 75°$?

(a) 0 (b) $\frac{13}{2}$ (c) 6 (d) $\frac{11}{2}$

13. What is the value of
$\dfrac{5\sin 75°\sin 77° + 2\cos 13°\cos 15°}{\cos 15°\sin 77°} - \dfrac{7\sin 81°}{\cos 9°}$?

(a) -1 (b) 0 (c) 1 (d) 2

14. What is $\log(\tan 1°) + \log(\tan 2°) + \log(\tan 3°) + \cdots + \log(\tan 89°)$ equal to?

(a) 0 (b) 1 (c) 2 (d) -1

15. If $\tan\theta + \dfrac{1}{\tan\theta} = 2$, then the value of $\tan^2\theta + \dfrac{1}{\tan^2\theta}$ is equal to

(a) 6 (b) 4 (c) 2 (d) 3

16. If $\sec\theta = \dfrac{13}{5}$, then $\dfrac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$ is equal to

(a) $1/3$ (b) -3 (c) 3 (d) $\sqrt{3}$

17. $\sin(45° + A) - \cos(45° - A)$ is equal to

(a) $1/\sqrt{2}$ (b) 1 (c) 0 (d) $\sqrt{2}$

18. What is $\sqrt{\dfrac{1+\sin\theta}{1-\sin\theta}}$ equal to?

(a) $\sec\theta - \tan\theta$ (b) $\sec\theta + \tan\theta$
(c) $\operatorname{cosec}\theta + \cot\theta$ (d) $\operatorname{cosec}\theta - \cot\theta$

19. $\sin(n+1)A\sin(n-1)A + \cos(n+1)A\cos(n-1)A$ is equal to

(a) $\sin 2A$ (b) $\cos 2A$ (c) $\tan 2A$ (d) $\cot 2A$

20. The value of $\dfrac{\cos 11° + \sin 11°}{\cos 11° - \sin 11°}$ is

(a) $\tan 56°$ (b) $\tan 34°$ (c) $\cot 56°$ (d) $\tan 11°$

21. If $\sin\theta = \dfrac{a^2 - b^2}{a^2 + b^2}$, then $\tan\theta$ is equal to

(a) $\frac{a^2+b^2}{2ab}$ (b) $\frac{2ab}{a^2-b^2}$ (c) $\frac{a^2-b^2}{2ab}$ (d) $\frac{ab}{a^2+b^2}$

22. If $\cos\theta = 0.96$, then $\dfrac{1}{\sin\theta} + \dfrac{1}{\tan\theta}$ is equal to

(a) 0.98 (b) 3 (c) 4 (d) 7

23. If $0 < x < 45°$ and $45° < y < 90°$, then which one of the following is correct?

(a) $\sin x = \sin y$ (b) $\sin x < \sin y$
(c) $\sin x > \sin y$ (d) $\sin x \le \sin y$

24. For what value of θ is $(\sin\theta + \operatorname{cosec}\theta) = 2.5$, where $0 < \theta \le 90°$?

(a) 30° (b) 45° (c) 60° (d) 90°

25. If $\dfrac{\cos x}{\cos y} = n$ and $\dfrac{\sin x}{\sin y} = m$, then $(m^2 - n^2)\sin^2 y$ is equal to

(a) $1 - n^2$ (b) $1 + n^2$ (c) m^2 (d) n^2

26. If $p = \tan^2 x + \cot^2 x$, then which one of the following is correct?

(a) $p \le 2$ (b) $p \ge 2$ (c) $p < 2$ (d) $p > 2$

27. The difference of the two angles in degree measure is 1 and their sum in circular measure is also 1. What are the angles in circular measure?

(a) $\left(\frac{1}{2} - \frac{\pi}{360}\right), \left(\frac{1}{2} + \frac{\pi}{360}\right)$ (b) $\left(\frac{1}{2} - \frac{90}{\pi}\right), \left(\frac{1}{2} + \frac{90}{\pi}\right)$

(c) $\left(\frac{1}{2} - \frac{\pi}{180}\right), \left(\frac{1}{2} + \frac{\pi}{180}\right)$ (d) None of these

28. What is the value of $[(1 - \sin^2\theta)\sec^2\theta + \tan^2\theta](\cos^2\theta + 1)$ when $0 < \theta < 90°$?

(a) 2 (b) > 2 (c) ≥ 2 (d) < 2

29. If $7\cos^2\theta + 3\sin^2\theta = 4$ and $0 < \theta < \dfrac{\pi}{2}$, then what is the value of $\tan\theta$?

(a) $\sqrt{7}$ (b) $\frac{7}{3}$ (c) 3 (d) $\sqrt{3}$

30. If $\cos 1° = p$ and $\cos 89° = q$, then which one of the following is correct?

(a) p is close to 0 and q is close to 1
(b) $p < q$
(c) $p = q$
(d) p is close to 1 and q is close to 0

31. If $0 \le \theta < \dfrac{\pi}{2}$ and $p = \sec^2\theta$, then which one of the following is correct?

(a) $p < 1$ (b) $p = 1$ (c) $p > 1$ (d) $p \ge 1$

32. What is the value of $\cos 1° \cos 2° \cos 3° \ldots \cos 90°$?

(a) $\frac{1}{2}$ (b) 0 (c) 1 (d) 2

33. If $\tan A = \dfrac{1 - \cos B}{\sin B}$, then what is $\dfrac{2\tan A}{1 - \tan^2 A}$ equal to?

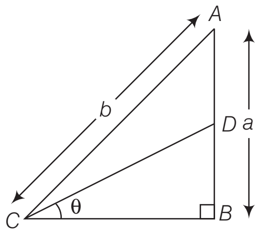
(a) $\frac{\tan B}{2}$ (b) $2\tan B$ (c) $\tan B$ (d) $4\tan B$

34. What is cot 15° cot 20° cot 70° cot 75° equal to?
(a) – 1 (b) 0 (c) 1 (d) 2

35. If $\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}=2$ with $0<\theta<90°$, then what is θ equal to?
(a) 30° (b) 45° (c) 60° (d) 75°

36. In a right ΔABC, right angled at B, the ratio of AB to AC is $1:\sqrt{2}$, then $\frac{2\tan A}{1-\tan^2 A}$ is equal to
(a) 2 (b) 1 (c) 3 (d) undefined

37. In figure, $AD = DB$, $\angle B = 90°$, then $\cos\theta$ is equal to



(a) $\frac{2\sqrt{b^2-a^2}}{\sqrt{4b^2-3a^2}}$ (b) $\frac{a}{\sqrt{4b^2-3a^2}}$
(c) $\frac{\sqrt{4b^2-3a^2}}{2}$ (d) None of these

38. If $0°\le x\le 90°$ and $\sin x+\sqrt{3}\cos x=1$, then what is the value of x?
(a) 30° (b) 45° (c) 60° (d) 90°

39. If $\tan A=\sqrt{2}-1$, then the value of $\operatorname{cosec} A\cdot\sec A$ is equal to
(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{2}{\sqrt{2}}$ (c) $2\sqrt{2}$ (d) $\frac{\sqrt{3}}{2}$

40. $\tan\theta=3$ and θ lies in third quadrant, then the value of $\sin\theta$ is
(a) $\frac{1}{\sqrt{10}}$ (b) $-\frac{1}{\sqrt{10}}$ (c) $\frac{-3}{\sqrt{10}}$ (d) $\frac{3}{\sqrt{10}}$

41. The value of (sin 20° cos 70° + cos 20° sin 70°) is
(a) 1 (b) 0 (c) – 1 (d) $\frac{1}{2}$

42. If ϕ and θ are supplementary angles, then
(a) $\sin\theta=\sin\phi$ (b) $\cos\theta=\cos\phi$
(c) $\tan\theta=\tan\phi$ (d) $\sec\phi=\operatorname{cosec}\theta$

43. If $x+y=90°$, then what is the value of $\sqrt{\cos x\operatorname{cosec} y-\cos x\sin y}$?
(a) $\cos x$ (b) $\sin x$ (c) $\sqrt{\cos x}$ (d) $\sqrt{\sin x}$

44. The value of $\sqrt{2+\sqrt{2+2\cos 4\theta}}$ is
(a) $2\cos^2\theta$ (b) $2\cos 2\theta$ (c) $2\cos\theta$ (d) $2\cos\frac{\theta}{2}$

45. If $\tan A=\frac{5}{6}$ and $\tan B=\frac{1}{11}$, then $A+B$ is equal to
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{3\pi}{2}$ (d) π

46. The value of $\frac{1}{\sqrt{2}}(\cos A-\sin A)$ is
(a) $\cos\left(\frac{\pi}{3}+A\right)$ (b) $\cos\left(\frac{\pi}{2}+A\right)$
(c) $\cos\left(\frac{\pi}{4}+A\right)$ (d) $\sin\left(\frac{\pi}{4}+A\right)$

47. $\sin\frac{\pi}{4}\cos\frac{\pi}{12}-\cos\frac{\pi}{4}\sin\frac{\pi}{12}$ is equal to
(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$

48. If $\tan\theta=\frac{x\sin\phi}{1-x\cos\phi}$ and $\tan\phi=\frac{y\sin\theta}{1-y\cos\theta}$, then $\frac{x}{y}$ is equal to
(a) $\frac{\sin\phi}{\sin\theta}$ (b) $\frac{\sin\phi}{\cos\theta}$ (c) $\frac{\sin\theta}{\sin\phi}$ (d) $\frac{\sin\theta}{1-\cos\phi}$

49. If $\sin(\theta+\phi)=2\sin(\theta-\phi)$, then
(a) $\cot\phi=3\tan\theta$ (b) $\tan\theta=3\tan\phi$
(c) $\sin\theta=3\sin\phi$ (d) $\sin\phi=\sin 2\theta$

50. If $\sin(\theta+\alpha)=\cos(\theta+\alpha)$, then $\tan\theta$ is equal to
(a) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (b) $\frac{\tan\alpha+1}{\tan\alpha-1}$ (c) $\frac{1-\cot\alpha}{1+\cot\alpha}$ (d) $\frac{\sin 2(\theta+\alpha)}{\cos(\theta+\alpha)}$

51. $\cos^2\frac{\theta-\phi}{2}-\sin^2\frac{\theta+\phi}{2}$ is equal to
(a) $\cos\phi\sin\theta$ (b) $\cos 2\theta\sin\phi$
(c) $\cos\theta\cos\phi$ (d) $\sin\theta\sin\phi$

52. If $x=r\sin\theta\cos\phi$, $y=r\sin\theta\sin\phi$, $z=r\cos\theta$, then $x^2+y^2+z^2$ is equal to
(a) $r^2\cos^2\phi$ (b) $r^2\sin^2\theta+r^2\cos^2\phi$
(c) r^2 (d) $\frac{1}{r^2}$

53. If $\cot\theta=\frac{7}{24}$ and $\pi<\theta<\frac{3\pi}{2}$, then the value of $\cos\theta-\sin\theta$ is
(a) $\frac{19}{25}$ (b) $\frac{18}{25}$
(c) $\frac{17}{25}$ (d) $\frac{18}{25}$

54. If $\alpha+\beta=\frac{\pi}{2}$ and $\sin\alpha=\frac{1}{2}$, then β is
(a) 30° (b) 60° (c) 45° (d) $22\frac{1°}{2}$

55. If $\sin x-\cos x=0$, then what is the value of $\sin^4 x+\cos^4 x$?
(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

56. If $3\cos x = 5\sin x$, then the value of $\dfrac{5\sin x - 2\sec^3 x + 2\cos x}{5\sin x + 2\sec^3 x - 2\cos x}$ is

(a) $\dfrac{361}{2397}$ (b) $\dfrac{271}{979}$ (c) $\dfrac{541}{979}$ (d) $\dfrac{127}{979}$

57. $\sqrt{\dfrac{1+\cos x}{1-\cos x}}$ is equal to

(a) $\sec x + \tan x$ (b) $\operatorname{cosec} x + \cot x$
(c) $\sec x - \tan x$ (d) $\operatorname{cosec} x - \cot x$

58. $\cos^4 x - \sin^4 x$ is equal to

(a) $2\cos^2 x - 1$ (b) $2\sin^2 x - 1$
(c) $\sin^2\theta - \cos^2\theta$ (d) None of these

59. $\dfrac{\sin\theta}{1+\cos\theta}$ is equal to

(a) $\dfrac{1-\cos\theta}{\sin\theta}$ (b) $\dfrac{\sin\theta}{\cos\theta}$ (c) $\dfrac{\cos\theta-1}{\sin\theta}$ (d) $\dfrac{1+\sin\theta}{\cos\theta}$

60. If $0 \le \theta \le 90°$, then $\left(\dfrac{5\cos\theta-4}{3-5\sin\theta} - \dfrac{3+5\sin\theta}{4+5\cos\theta}\right)$ is equal to

(a) 0 (b) 1 (c) $\dfrac{1}{4}$ (d) $\dfrac{1}{2}$

61. If $\cos(\alpha+\beta) = \dfrac{4}{5}$ and $\sin(\alpha-\beta) = \dfrac{5}{13}, \alpha, \beta$ lies between 0 and $\dfrac{\pi}{4}$, then the value of $\tan 2\alpha$ is

(a) $\dfrac{56}{33}$ (b) $\dfrac{56}{23}$ (c) $\dfrac{43}{33}$ (d) $\dfrac{34}{33}$

62. $\dfrac{\tan(45°+x)}{\tan(45°-x)}$ is equal to

(a) $\left(\dfrac{1+\tan x}{1-\tan x}\right)^2$ (b) $\left(\dfrac{1-\tan x}{1+\tan x}\right)^2$
(c) $\dfrac{2\sin^2\frac{x}{2}}{1+\cos^2\frac{x}{2}}$ (d) None of these

63. What is the simplest value of $\dfrac{(1-\sin A\cos A)(\sin^2 A - \cos^2 A)}{\cos A(\sec A - \operatorname{cosec} A)(\sin^3 A + \cos^3 A)}$?

(a) $\sin A$ (b) $\cos A$ (c) $\sec A$ (d) $\operatorname{cosec} A$

64. If $A + B = 45°$, then $(1+\tan A)(1+\tan B)$ is equal to

(a) 1 (b) -1 (c) 2 (d) -2

65. $\left(\dfrac{\cos 2B - \cos 2A}{\sin 2B + \sin 2A}\right)$ is equal to

(a) $\tan(A-B)$ (b) $\tan(A+B)$ (c) $\cot(A-B)$ (d) $\cot(A+B)$

66. $\sec^2 x + \tan^2 x = 7$, the value of x is

(a) 15° (b) 30° (c) 45° (d) 60°

67. $\cos 3\theta + \sin 3\theta$ is maximum, when θ is

(a) 15° (b) 30° (c) 45° (d) 60°

68. If $\sin^2 x + \sin^2 y + \sin^2 z = (\sin x + \sin y + \sin z)^2$, then which of the following expressions must necessarily vanish?

(a) $\tan x + \tan y + \tan z$ (b) $\cos x + \cos y + \cos z$
(c) $\dfrac{1}{\sin x} + \dfrac{1}{\sin y} + \dfrac{1}{\sin z}$ (d) $\dfrac{1}{\cos x} + \dfrac{1}{\cos y} + \dfrac{1}{\cos z}$

69. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is equal to

(a) -1 (b) 2 (c) -2 (d) 1

70. $\cos 35° + \cos 85° + \cos 155°$ is

(a) 0 (b) $\dfrac{1}{\sqrt{3}}$ (c) $\dfrac{1}{\sqrt{2}}$ (d) $\cos 275°$

71. $\sin 12° \sin 24° \sin 48° \sin 84°$ is

(a) $\dfrac{1}{16}$ (b) $\dfrac{3}{64}$ (c) $\dfrac{3}{15}$ (d) 0

72. If $\tan\theta = \dfrac{p}{q}$, then the value of $p\cos 2\theta + q\sin 2\theta$ is

(a) p (b) q
(c) $\dfrac{q(3q^2-p^2)}{p^2+q^2}$ (d) $\dfrac{p(3q^2-p^2)}{p^2+q^2}$

73. If $x = y\cos\dfrac{2\pi}{3} = z\cos\dfrac{4\pi}{3}$, then $xy + yz + zx$ is equal to

(a) 1 (b) -1 (c) 0 (d) 2

74. If $\sin\theta + \cos\theta = m$ and $\sec\theta + \operatorname{cosec}\theta = n$, then $n(m^2-1)$ is equal to

(a) $\dfrac{1}{2m}$ (b) m (c) $2m$ (d) $\dfrac{m}{2}$

75. If $\operatorname{cosec}\theta - \sin\theta = a^3$, $\sec\theta - \cos\theta = b^3$, then $a^2b^2(a^2+b^2)$ is equal to

(a) -1 (b) 1 (c) 2 (d) -2

76. The maximum value and minimum value of $(1+\cos 2x)$ are

(a) -1 and 1 (b) 1 and 2 (c) $\dfrac{-1}{2}$ and $\dfrac{1}{2}$ (d) 0 and 2

77. If $\tan\theta + \tan\phi = a$ and $\cot\theta + \cot\phi = b$, then $\cot(\theta+\phi)$ is equal to

(a) $\dfrac{1}{a} + \dfrac{1}{b}$ (b) $\dfrac{1}{a} - \dfrac{1}{b}$
(c) $a - b$ (d) $a + b$

78. $\tan 75° - \tan 30° - \tan 75° \tan 30°$ is equal to

(a) -1 (b) 1 (c) 0 (d) 2

79. The value of $\sin^2(15°+A) - \sin^2(15°-A)$ is equal to

(a) $\dfrac{1}{2}\cos 2A$ (b) $\dfrac{1}{2}\sin 2A$ (c) $\dfrac{1}{2}\tan 2A$ (d) $\cot 2A$

80. The value of $\dfrac{\sin 38° - \cos 68°}{\cos 68° + \sin 38°}$ is equal to

(a) $\sqrt{3} \tan 8°$ (b) $\sqrt{3} \cot 8°$
(c) $\sqrt{3} \sin 38°$ (d) $\sqrt{3} \sin 8°$

81. $2 \cos x - \cos 3x - \cos 5x$ is equal to

(a) $8 \cos^3 x \sin^2 x$ (b) $12 \cos^2 x \sin^3 x$
(c) $16 \cos^3 x \sin^2 x$ (d) $32 \cos^3 x \sin x$

82. $\cos 4x$ is equal to

(a) $1 + 2 \sin^2 2x$ (b) $2 \cos^2 2x$
(c) $1 - 8 \sin^2 x \cos^2 x$ (d) $1 + 8 \sin^2 x \cos^2 x$

83. $2 \sin A \cos^3 A - 2 \sin^3 A \cos A$ is equal to

(a) $\frac{1}{2} \sin 4A$ (b) $\frac{1}{2} \cos 4A$ (c) $\frac{1}{2} \tan 4A$ (d) $\frac{1}{2} \cot 4A$

84. If $a \cos \theta - b \sin \theta = c$, then what is the value of $a \sin \theta + b \cos \theta$?

(a) $\pm \sqrt{a^2 + b^2 + c^2}$ (b) $\pm \sqrt{a^2 - b^2 + c^2}$
(c) $\pm \sqrt{a^2 + b^2 - c^2}$ (d) $\pm \sqrt{a^2 - b^2 - c^2}$

85. If $\cos x + \cos^2 x = 1$, then the value of $\sin^8 x + 2 \sin^6 x + \sin^4 x$ is

(a) 0 (b) -1 (c) 2 (d) 1

86. If $m = \operatorname{cosec} x - \sin x$ and $n = \sec x - \cos x$, then $\tan x$ is equal to

(a) $\left(\frac{n}{m}\right)^{2/3}$ (b) $\left(\frac{n}{m}\right)^{1/3}$
(c) $\left(\frac{n}{m}\right)$ (d) $\left(\frac{n}{m}\right)^2$

87. If $2 \cos \theta = x + \dfrac{1}{x}$, then $2 \cos 3\theta$ is equal to

(a) $x^3 + \frac{1}{x^3}$ (b) $x^2 + \frac{1}{x^2}$ (c) $\frac{x^2}{1 + x^2}$ (d) None of these

88. If $\cos \theta \ge \dfrac{1}{2}$ in the first quadrant, then which one of the following is correct?

(a) $\theta \le \frac{\pi}{3}$ (b) $\theta \ge \frac{\pi}{3}$ (c) $\theta \le \frac{\pi}{6}$ (d) $\theta \ge \frac{\pi}{6}$

89. If $\sin A = \dfrac{3}{5}$ and $\cos B = \dfrac{12}{13}$, then the value of $\dfrac{\tan A - \tan B}{1 + \tan A \tan B}$ is equal to

(a) $\frac{23}{16}$ (b) $\frac{16}{63}$ (c) $\frac{1}{63}$ (d) $\frac{13}{63}$

90. If $B + C = 60°$, then which of the following is correct statement?

(a) $\sin(120° - B) = \sin(120° - C)$
(b) $\sin(120° + B) = \sin(180° + C)$
(c) $\cos(120° - B) = \sin(120° + C)$
(d) $\tan(B) = \tan(120° + C)$

91. What is the value of x in the equation

$$x \frac{\operatorname{cosec}^2 30° \sec^2 45°}{8 \cos^2 45° \sin^2 60°} = \tan^2 60° - \tan^2 30° \ ?$$

(a) $x = 1$ (b) $x = 2$ (c) $x = \frac{1}{2}$ (d) $x = \frac{3}{2}$

92. Under which one of the following conditions is the trigonometrical identity $\sin x / (1 + \cos x) = (1 - \cos x) / \sin x$ true?

(a) x is not a multiple of 360°
(b) x is not an odd multiple of 180°
(c) x is not a multiple of 180°
(d) None of the above

93. If $\cos x = k \cos(x - 2y)$, then $\tan(x - y) \tan y$ is equal to

(a) $\frac{1 + k}{1 - k}$ (b) $\frac{1 - k}{1 + k}$ (c) $\frac{2k}{k + 1}$ (d) $\frac{k - 1}{2k + 1}$

94. The value of $\dfrac{\sin(x + y) - 2 \sin x + \sin(x - y)}{\cos(x + y) - 2 \cos x + \cos(x - y)}$ is

(a) $\cot x$ (b) $\tan x$ (c) $\sin x$ (d) $\operatorname{cosec} x$

95. If $\sin x + \sin y = a$ and $\cos x + \cos y = b$, then $\tan \dfrac{x + y}{2}$ is

(a) $\frac{4}{a^2 + b^2}$ (b) $\frac{b}{a}$ (c) $\frac{a}{b}$ (d) $\frac{4}{a^2 - b^2}$

96. The largest hand of a clock is 42 cm long, then the distance covered by the extremity in 20 min is

(a) 88 cm (b) 80 cm (c) 82 cm (d) 84 cm

97. If $0° < x < 90°$ and $2 \sin x + 15 \cos^2 x = 7$, then what is the value of $\tan x$?

(a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) $\frac{3}{4}$ (d) $\frac{4}{3}$

98. If ABC is a triangle and $A + B + C = 180°$, then $\tan A + \tan B + \tan C$ is

(a) $\tan A \tan B \tan C$ (b) $\tan \frac{A + B}{2} + \tan \frac{B + C}{2}$
(c) $\frac{\tan A + \tan B}{\tan C}$ (d) $\cot A \cot B \cot C$

99. If $x = a \sec \theta \cos \phi$, $y = b \sec \theta \sin \phi$ and $z = c \tan \theta$, then the value of $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2}$ is

(a) 9 (b) 0 (c) 1 (d) 4

100. Which one of the following statements is correct?

(a) The squares of the tangents of the angles 30°, 45°, 60° are in G.P.
(b) The squares of the sines of the angles 30°, 45°, 60° are in G.P.
(c) The squares of the secants of the angles 30°, 45°, 60° are in A.P.
(d) The squares of the tangents of the angles 30°, 45°, 60° are in A.P.

101. If A, B, C and D are the successive angles of a cyclic quadrilateral, then what is $\cos A + \cos B + \cos C + \cos D$ are equal to?

(a) 4 (b) 2 (c) 1 (d) 0

102. Which one of the following is correct?

(a) There is only one θ with $0° < \theta < 90°$ such that $\sin\theta = a$, where a is a real number
(b) There is more than one θ with $0° < \theta < 90°$ such that $\sin\theta = a$, where a is a real number
(c) There is no θ with $0° < \theta < 90°$ such that $\sin\theta = a$, where a is a real number
(d) There are exactly two θ's with $0° < \theta$ $90°$ such that $\sin\theta = a$, where a is a real number

103. If $\sin(B + C - A) = \cos(C + A - B) = \tan(A + B - C) = 1$, then the angles A, B, C which are positive acute angles are, respectively

(a) 45°, 80°, 105° (b) $22\frac{1°}{2}, 67\frac{1°}{2}$, 45°
(c) 20°, 70°, 90° (d) 30°, 60°, 90°

104. If A, B, C and D be the angles of a cyclic quadrilateral taken in order, then $\cos(180° + A) + \cos(180° + B) + \cos(180° + C) + \cos(180° + D)$ is equal to

(a) $2(\cos A + \cos B)$ (b) $2(\cos A + \cos D)$
(c) 0 (d) $4\cos A$

105. If ABC is a right angled triangle at C and having u units, v units and w units as the lengths of its sides opposite to be vertices A, B and C respectively, then what is $\tan A + \tan B$ equal to?

(a) $\frac{u^2}{vw}$ (b) 1 (c) $u + v$ (d) $\frac{w^2}{uv}$

106. The Earth takes 24 h to rotate about its own axis. Through what angle will it turn in 4 h and 12 min?

(a) 63° (b) 64° (c) 65° (d) 70°

107. Assume the Earth to be a sphere of radius R. What is the radius of the circle of latitude 40°S?

(a) $R\cos 40°$ (b) $R\sin 80°$
(c) $R\sin 40°$ (d) $R\tan 40°$

108. If $\sin 3\theta = \cos(\theta - 2°)$, where 3θ and $(\theta - 2°)$ are acute angles, then what is the value of θ?

(a) 22° (b) 23°
(c) 24° (d) 25°

109. Consider the following statements:

I. $\frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin^2\theta} = \cos^2\theta\,(1 + \tan\theta)(1 - \tan\theta)$

II. $\frac{1 + \sin\theta}{1 - \sin\theta} = (\tan\theta + \sec\theta)^2$

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

110. If $A = \frac{\pi}{6}$ and $B = \frac{\pi}{3}$, then which of the following is/are correct?

I. $\sin A + \sin B = \cos A + \cos B$
II. $\tan A + \tan B = \cot A + \cot B$

Select the correct answer using the codes given below

(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II

111. For any quadrilateral $ABCD$ which of the following statements are true?

I. $\sin(A + B) + \sin(C + D) = 0$
II. $\cos(A + B) + \cos(C + D) = 0$

Select the correct answer using the codes given below

(a) Only I (b) Only II
(c) Both I and II (d) None of these

112. Consider the following statements

I. 1° in radian measure is less than 0.03 radians.
II. 1 radian in degree measure is greater than 45°.

Which of the above statement is/are correct?

(a) Only II (b) Only I
(c) Neither I nor II (d) Both I of II

113. Consider the following statements

I. $\sin\theta\cdot\sin(60°+\theta)\cdot\sin(60°-\theta) = \frac{1}{4}\sin 3\theta$

II. $\cos\theta\cdot\sin(30°+\theta)\cdot\sin(30°-\theta) = \frac{1}{4}\cos 3\theta$

III. $\sin\theta\cdot\cos(30°+\theta)\cdot\cos(30°-\theta) = \frac{1}{4}\sin 3\theta$

Which of the above statement is/are correct?

(a) Only I
(b) Only II
(c) Only III
(d) All of the above

114. If ABC is a right angled triangle, then which of the following statements are correct?

I. $\sin(A + B) = \sin C$

II. $\sin\left(\frac{A + B}{2}\right) = \cos\frac{C}{2}$

III. $\tan\frac{(A + B - C)}{2} = \cot C$

IV. $\tan\frac{(A - B - C)}{2} = -\cot A$

Select the correct answer using the codes given below

(a) I and II (b) I, II and III
(c) I, II and IV (d) All of these